Supplementary material

S1. Evidence classification

Each parameter is labeled OBSERVED, SOURCED, CALCULATED, or ASSUMED_SCENARIO. The source, assumption, acquisition, and data dictionaries distributed with the repository provide field-level provenance. Scenario outputs do not become empirical evidence through simulation.

S2. Analytical break-even identities

For width reduction with unchanged basis mass and process yield, material ratio is the product of width and successful-event length ratios. The exact length break-even is the inverse width ratio and the constant-elasticity break-even is minus one. For straw shortening with zero shifted burden, the expected additional replacement-rate break-even is the reduction divided by retained length. Under an independent per-attempt failure model, expected attempts are one divided by success probability and the zero-shift failure boundary equals the length-reduction fraction. Shifted container burden is defined per successful serving and added once. Center-perforation benefit is calculated from a task mixture and expected half-units, not assumed to be 50%.

S3. Scenario distributions


```
{
  "seed": 20260924,
  "monte_carlo_iterations": 200000,
  "toilet_width": {
    "baseline_width_mm": 114.0,
    "candidate_widths_mm": [100.0, 95.0, 90.0, 80.0],
    "primary_candidate_width_mm": 95.0,
    "primary_candidate_status": "ILLUSTRATIVE_REFERENCE_SCENARIO",
    "elasticity_triangular": [-1.2, -0.35, 0.15],
    "success_ratio_triangular": [0.95, 0.99, 1.0],
    "manufacturing_penalty_uniform": [0.0, 0.02]
  },
```

```
  "center_perforation": {
    "half_suitable_probability_beta": [3.0, 3.0],
    "half_task_unit_multiplier_triangular": [1.0, 1.15, 1.8],
    "success_ratio_triangular": [0.96, 0.995, 1.0],
    "manufacturing_penalty_uniform": [0.0, 0.02],
    "reported_task_mix_scenarios": [0.25, 0.5, 0.75],
    "reported_half_task_multiplier": 1.2
  },
  "straw": {
    "primary_length_reduction_fraction": 0.2,
    "monte_carlo_replacement_formulation": "EXPECTED_ADDITIONAL_REPLACEMENT_RATE",
    "expected_additional_replacement_rate_triangular": [0.0, 0.05, 0.35],
    "shifted_container_burden_uniform": [0.0, 0.05],
    "reported_length_reductions": [0.1, 0.2, 0.3],
    "reported_expected_additional_replacement_rates": [0.0, 0.1, 0.2, 0.3, 0.4],
    "reported_failure_probabilities": [0.0, 0.1, 0.2, 0.3, 0.4]
  },
  "logistics": {
    "illustrative_carton_internal_mm": [400.0, 300.0, 250.0],
    "illustrative_unit_other_dimensions_mm": [20.0, 20.0],
    "illustrative_straw_lengths_mm": [165.0, 132.0, 115.5],
    "vehicle_payload_kg": 24000.0,
    "vehicle_volume_m3": 82.0
  }
}
```

S4. Stress tests

The pipeline tests zero and strong compensation, zero logistics benefit, unchanged packaging, integer-packing inefficiency, process penalties, low acceptance, high replacement, alternative functional units, conservative functional minima, and removal of PFAS evidence. The one-way sensitivity table and full fixed-seed draws are provided as machine-readable files.

S5. Proposed toilet-paper validation

Use a randomized balanced crossover of baseline and candidate geometry. Measure roll mass or instrumented dispensing at the event level. Prespecify successful-event material as primary outcome and record length, units, folding, repeat use, adequacy, failure, and acceptability. Cross width and center perforation in a 2×2 design and measure selection and cross-tear errors. Ethics review is required before human enrollment.

S6. Proposed straw validation

Use a geometric bench rig varying container height, fill level, insertion point, angle, beverage viscosity, and straw length. Record accessible liquid, residual mass, protrusion, handling force, buckling, leakage, detachment, and replacement. Only candidates meeting bench criteria should proceed to randomized user evaluation.

S7. Reproduction

Create a Python environment from requirements.txt and run `make all`. The build regenerates calculated data, figures, tables, manuscript, submission files, and quality-control reports. Fixed random seeds are stored in config/scenarios.json.